

Since publishing that article, I have come to realize that calibrating \$Y to *wealth* instead of *spending* will work better. Attempting to calibrate downside spending can easily create a situation in which no spending rule works, which was an issue in my published article. Higher spending rates caused spending to fall too far, while lower spending rates impeded spending from rising enough. If the initial spending rate is too high, spending will plummet too quickly without recovery on the downside, and the \$Y spending threshold cannot be reached. At the same time, if spending is too low initially, it may never rise sufficiently in the bad luck scenarios to meet the needed threshold. There may not be any level of initial spending that can work. A happy medium was not always available, but defining \$Y as wealth instead of spending corrects for this.

As well, a number of spending rules include sudden and large discrete changes to spending throughout retirement. With an approach like the traditional 4 percent rule, when the portfolio is depleted, someone starting with \$1 million may spend \$40,000 one year and \$0 the next. This can make it harder to compare different spending strategies, since the PAY Rule may calibrate to a spending strategy that changes dramatically in the year following Z. By shifting the focus to remaining wealth, we also reduce the incidence of these discrete shifts to spending.

The PAY Rule works very well when defined as follows:

PAY Rule™ to Calibrate Retirement Spending

Retiree accepts an X% Probability that wealth falls below a threshold Amount of \$Y (in inflation-adjusted terms) by Year Z of retirement (Probability, Amount, Year).

The PAY Rule acts as a behind-the-scenes constraint to calibrate shortfalls between the strategies, allowing for a better overall understanding of the distribution of potential spending afforded by the strategies.

An important remaining issue, though, is how individuals decide on the parameters for the PAY Rule. These three parameters should be considered:

1. Length of the planning horizon,